Workshop 3: Sentiment Analysis in Python

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Goals of Today's Workshop

01

Run pretrained classifier

Execute emotion analysis model in Google Colab environment

02

Generate results file

Create output with emotion labels and confidence scores

03

Visualise emotion distribution

Compare emotional patterns between two video comment datasets

What We're NOT Doing

No model training today—we're leveraging pretrained capabilities

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Dataset Overview

Dataset Details

Cleaned YouTube comments from **Islam, Judaism, and Christianity – A Conversation**

https://www.youtube.com/watch?v=INIG636SnU4

Ethics

- Educational use only
- Text-only analysis (no user profiling)
- Do not share raw dataset outside workshop

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Session Roadmap

1. Core concepts & reading outputs
2. Hands-on coding in Colab (Steps 1–4)
3. Wrap-up, limitations & Q&A

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Sentiment Analysis

Today’s roadmap

- Quick warm-up
- A short audience activity
- What sentiment analysis is
- Where it is used
- Why it matters for Humanities
- Hands-on: Hugging Face + Google Colab

How are you feeling today?

Describe your current mood in 1–3 words on Mentimeter

(anonymous, no personal details)

https://www.menti.com/albz7mo7fchd

Activity: Let’s do sentiment analysis

We will use 3 short responses from the audience.

For each text, we will apply three tasks:

1. Simplest task:

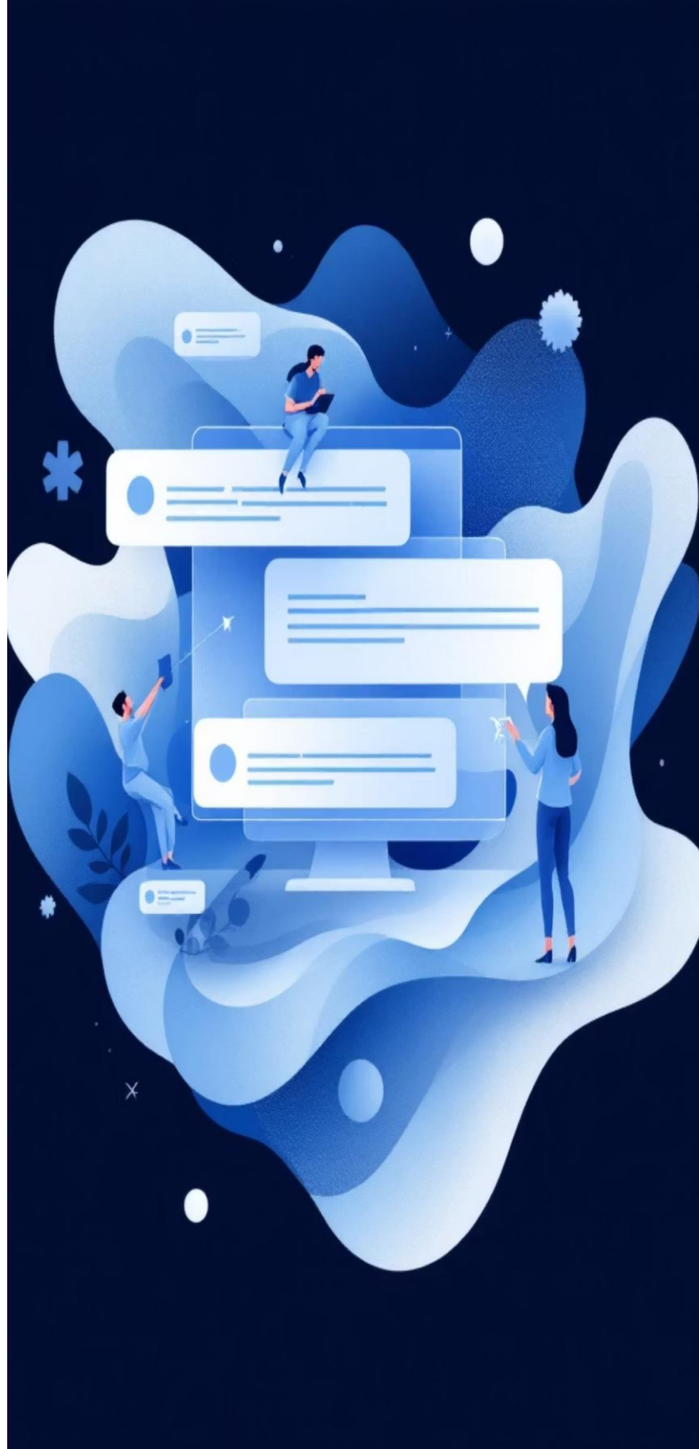
Judge if the sentiment of this text is positive or negative

2. More complex task:

Rank the sentiment of this text from 1 to 5

3. Advanced task:

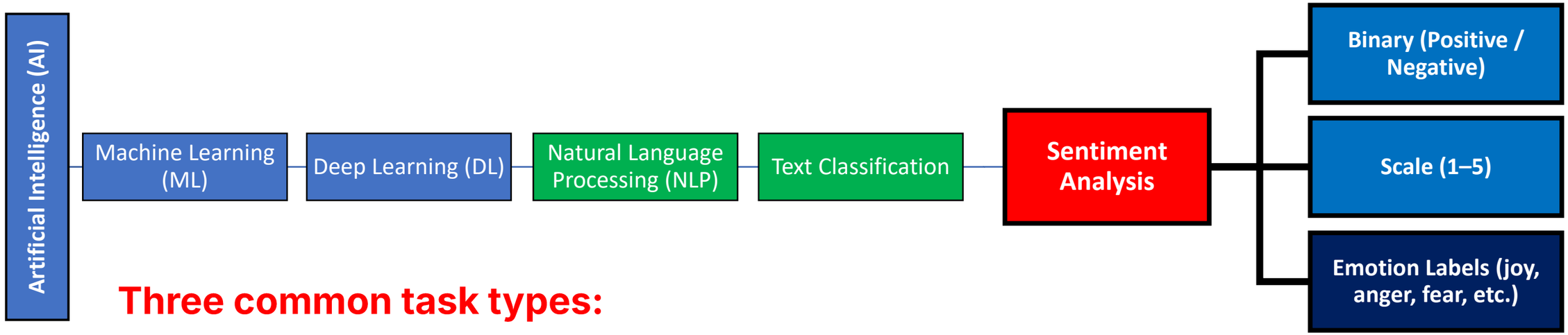
Label the sentiment of this text (Anger, Fear, Joy, Love, Sadness, Surprise, and Neutral)



What we just did: Sentiment Analysis

Sentiment analysis is an NLP task that classifies emotional tone in text.

Where it fits in AI



Three common task types:

- Positive / Negative (binary)
- 1-5 rating (scale)
- **Emotion labels (joy, anger, fear, etc.)**

Model output: label + score (confidence, not intensity)

Where is sentiment analysis used?

- Social media & public reaction tracking
- Reviews and customer feedback
- Media/marketing: compare audience reception across campaigns or over time

Why it matters for Humanities

- **Focus:** meaning, context, genre
- **Use cases:** reception/discourse analysis, narrative mood, cultural trends
- **Key cautions:** sarcasm, negation, mixed emotions, missing context
- Use the results as clues, not final answers

Hugging Face

- **AI community:**
  "The platform where the machine learning community collaborates on models, datasets, and applications"

- **Sign up: https://huggingface.co/**

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Hugging Face Models

Hugging Face | sentiment | Models | Datasets | Spaces | Community | Docs | Enterprise | Pricing

cardiffnlp/twitter-roberta-base-sentiment-latest | like 769 | Follow Cardiff NLP 239

Text Classification | Transformers | PyTorch | TensorFlow | tweet_eval | English | roberta | arxiv:2202.03829 | License: cc-by-4.0

Model card | Files and versions xet | Community 43 | Deploy | Use this model

Edit model card

Twitter-roBERTa-base for Sentiment Analysis - UPDATED (2022)

This is a RoBERTa-base model trained on ~124M tweets from January 2018 to December 2021, and finetuned for sentiment analysis with the TweetEval benchmark. The original Twitter-based RoBERTa model can be found here and the original reference paper is TweetEval. This model is suitable for English.

- Reference Paper: TimeLMs paper.
- Git Repo: TimeLMs official repository.

Labels: 0 -> Negative; 1 -> Neutral; 2 -> Positive

This sentiment analysis model has been integrated into TweetNLP. You can access the demo here.

Example Pipeline

```
from transformers import pipeline
sentiment_task = pipeline("sentiment-analysis", model=model_path, tokenizer=model
sentiment_task("Covid cases are increasing fast!")
```

```
[{'label': 'Negative', 'score': 0.7236}]
```

Downloads last month
4,788,686

Inference Providers NEW | HF Inference API

Text Classification | Examples

Your prompt here...

Generate

View Code Snippets | Maximize

Model tree for cardiffnlp/twitter-roberta-base-sentiment-latest

Adapters	10 models
Finetunes	222 models
Quantizations	1 model

Dataset used to train cardiffnlp/twitter-roberta-base-sentiment-latest

cardiffnlp/tweet_eval
Viewer • Updated Jan 4, 2024 • 201k • 22.8k • 136

Hands-On Setup (Step 0)

Open notebook

Launch provided Google Colab notebook

https://shorturl.at/xdtt9

Verify data

Confirm downloaded CSV file

https://shorturl.at/esGaC

Preview content

Check first 3–5 rows contain comment text and video identifiers

Load Libraries & Model (Step 1)

Step 1

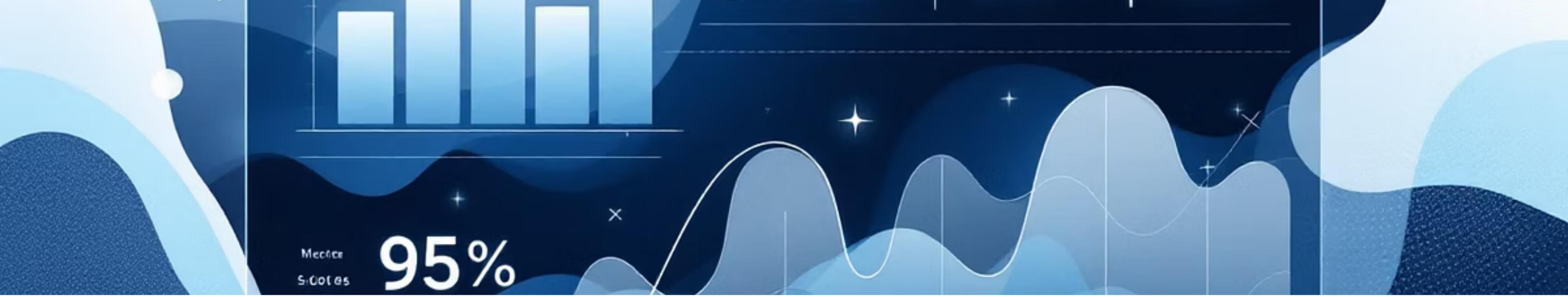
```
# @title Step 1

# Load modules
from transformers import pipeline
from matplotlib import pyplot as plt
import seaborn as sns
import csv
import pandas as pd

# Load the fine-tuned BERT Mini sentiment analysis model
sentiment_analysis = pipeline(
    "text-classification",
    model="j-hartmann/emotion-english-distilroberta-base",
    truncation=True,
    max_length=512
)
```

Run Analysis (Step 2)

Step 2

```
# @title Step 2

result = sentiment_analysis("I feel happy!")
print(result)
```

```
[{'label': 'joy', 'score': 0.9896815419197083}]
```



Interpreting Model Output

Label

Predicted emotion category (e.g., joy, anger, fear)

Score

Model's confidence or probability value

> **Important:** Score is not emotional intensity, not psychological diagnosis, not objective truth

Read data, run analysis, save output (Step 3)

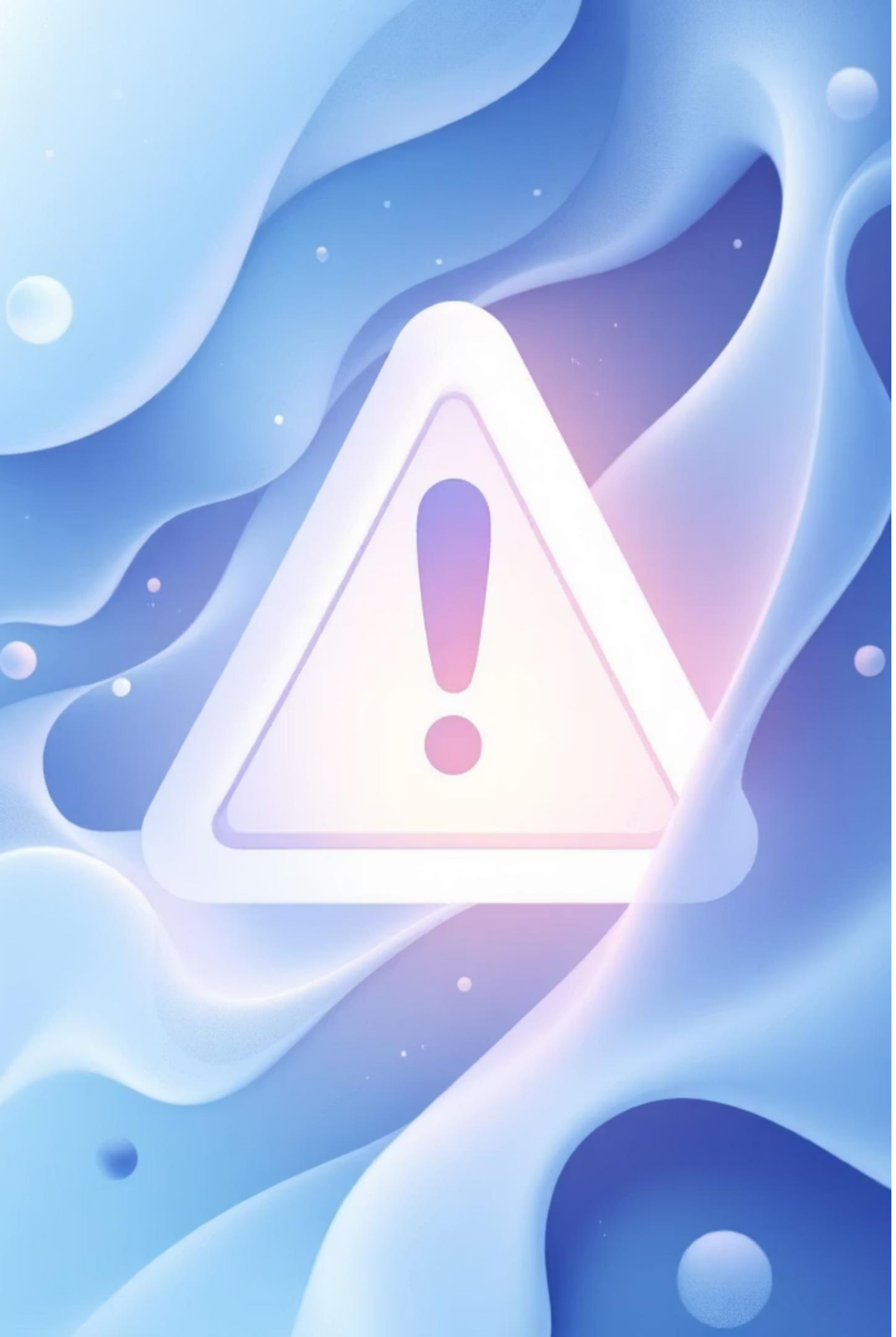
Step 3

```python
# @title Step 3

# Analyze sentiment for each row in your CSV file

results = []

with open('Day_1_workshop_3_sentiment_analysis_in_python.csv', mode='r', newline='', encoding='utf-8-sig') as file:
    reader = csv.reader(file)
    next(reader)  # skip the header

    # Iterate through each row in the CSV file
    for row in reader:

        # Apply sentiment analysis
        sentiment_result = sentiment_analysis(row[1])

        # Extract sentiment label
        sentiment_label = sentiment_result[0]['label']

        # Append the result as a dictionary
        results.append({'text': row[1], 'sentiment_label': sentiment_label, 'video': row[-1]})

        # Print results
        print('text:', row[1], 'sentiment_label:', sentiment_label)


# Create a DataFrame from the results
df = pd.DataFrame(results)

# Save the DataFrame to a single CSV file
df.to_csv("sentiment_results.csv", index=False, encoding="utf-8-sig")
```

Visualise Results (Step 4)

Step 4

```python
# @title Step 4

# Define the desired order of sentiment labels
sentiment_order = ['joy', 'love', 'surprise', 'fear', 'anger', 'sadness', 'neutral']

# Ensure the sentiment_label is a categorical type with the specified order
df['sentiment_label'] = pd.Categorical(df['sentiment_label'], categories=sentiment_order, ordered=True)

# Get unique videos
unique_videos = df['video'].unique()

# Set some styles
sns.set_style("whitegrid")
sns.set_context("talk")

# Create subplots
fig, axs = plt.subplots(len(unique_videos), 1, figsize=(8, 6 * len(unique_videos)), sharey=True)

# Loop through each unique video
for i, video in enumerate(unique_videos):
    df_video = df[df['video'] == video]
    sentiment_counts = df_video['sentiment_label'].value_counts().reindex(sentiment_order, fill_value=0)

    sns.barplot(x=sentiment_counts.values, y=sentiment_counts.index, ax=axs[i])

    axs[i].set_title(f'Distribution of Sentiment Labels for Video: {video}', fontsize=14, fontweight='bold')
    axs[i].set_xlabel('Count', fontsize=12)
    axs[i].set_ylabel('Sentiment Label', fontsize=12)
    axs[i].grid(axis='x', linestyle='--', alpha=0.7)

    for container in axs[i].containers:
        axs[i].bar_label(container, fmt='%d')

# Adjust layout
plt.tight_layout()
plt.show()
```



Common Limitations

Be careful of overinterpretation due to:

Sarcasm / Irony

Model struggles with tone reversal

Negation patterns

E.g., "not good" confuses simple classifiers

Mixed emotions

Single comment expresses multiple feelings

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Recap & Q&A

Workflow Recap

1

Load model

2

Test

3

Run on dataset

4

Visualise

Key Discussion

Do these labels reflect actual emotion or just stylistic tone?

Q&A

Questions and discussion

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